

Zero Emission Electri
Ferry at Varana

inland waterways with total estimated investment of around **Rs 4,800 crore** being executed by IWAI and state agencies

Urban water transportation through roll-on-roll-off (RORO)- (ROPAX) and Passenger Jetties: RORO and ROPAX service as a transport service has significant inherent benefits such as reduction in the travel time vis-à-vis land routes, lesser vehicular emission and easing of road traffic congestion. India, with a long coastline and riverine connections, has great potential for executing such projects. To leverage the low cost and environmental benefits of urban water transport vis-à-vis other modes such as road and rail, the government has identified more than **70 RoRo – RoPax & Passenger Jetty projects** across the country under Sagarmala Programme. Out of these, **9 projects have already been completed** with a total financial assistance of ~Rs 207 Cr.

Among these, one of the key impact project is the ROPAX ferry commenced in **Ghogha-Hazira** resulting in reduction in travel time from **10 hours to 4 hours** apart from facilitating quick movement of **15,000+ trucks, 50,000 cars and around 2 lakh passengers**. Similarly, to ease the road congestion in Mumbai, the government has launched ferry service between Mumbai-Mandwa resulting in reduction of distance of **109 km by road to 18.5 km sea-distance**, time reduction from **3 hours to just 45 minutes**. It has resulted in providing much needed relief to passengers resulting in movement of **5.5 lakh+ passengers and 1 lakh+ vehicles**.

GREEN SHIPPING

To reduce emissions from maritime sector in line with International Maritime Organization (IMO)'s Green House Gases (GHG) reduction strategy, government is working towards development of zero and low-emission solutions in the maritime sector. As per India's COP 26 goals, government has set climate targets limiting the global warming well below 2 degrees Celsius and efforts are being made to limit temperature increase to 1.5 degree Celsius above pre-industrial levels. Multiple initiatives have been already taken by the government for reducing maritime sector's carbon footprint. Some of the key measures taken in this regard include -increasing the share of solar & wind power, provision of shore power to port crafts/vessels and using electricity powered port equipment at the Major Ports. Further, the government has also drafted a "Green Port Policy" to suggest framework and guidelines for incorporation of green initiatives in the port sector. It captures the focused areas, measured outcomes, implementation roadmap and cost recovery mechanism for the port operators and port authorities which will help India in achieving the 'Intended Nationally Determined Contributions (INDCs)' target, as well as the International Maritime Organization (IMO) 2030 targets.

Moreover, Cochin Shipyard Limited (CSL) which is India's largest shipbuilding and maintenance facility under the government's aegis is spearheading various green vessel technology projects to promote Green Shipping in the country. These include development of **green urban mobility solutions** like **Hybrid Electric Ferries, autonomous Zero-emission vessels**, pilot project on **Hydrogen Fuel Cell Ferry, Electric Catamaran Water Taxi**, Hybrid Electric Ro-Ro, Hybrid LNG-Electric Inland Cargo Carrier, Hybrid Tugs, etc.

Possibilities are also being explored for deployment of **fully electric ferry and hydrogen fuel cell ferries at Inland Waterways** for enhancement of river cruise tourism in the country. For instance, Electric Catamaran Water Taxi deployment is being considered at Varanasi and Guwahati in first phase. Additionally, use of Hybrid Electric Roll on -Roll off (Ro-Ro) vessels having dual fuel system (LNG+ battery) is being considered at Guwahati for river crossings.

> DPA has emerged as a vibrant, world class multi-cargo port offering mutually beneficial and sustainable services to the Port Users and port based industries, thereby driving the economic growth of the region. The port is further planning to become a Mega Port and increase its capacity to more than 300 MMTPA for which two major projects for an estimated cost of about Rs.6000 crore are already initiated under PPP.
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> – S.K. Mehta, IFS, Chairman, Deendayal Port Authority


Liquid Terminal Facility at JNPA

PORT CONNECTIVITY

Port connectivity is one of the critical aspects for the overall maritime trade. The government through the Sagarmala programme has facilitated faster and efficient cargo movement between ports and their respective hinterland with an aim to minimise the overall logistics cost of the country. Timely implementation of port connectivity projects has enhanced overall freight efficiency and a testimony to government's efforts in this regard is successful completion of **69 port connectivity projects worth Rs. 32,000 Cr.** resulting in addition of **~1,300 km length** of road & rail connectivity to ports while another **67 projects of 5,500 km** length are under implementation.

PM GatiShakti
National Master Plan for Multi-Modal Connectivity

Under the visionary leadership of Hon'ble PM exhorting for seamless infrastructure planning and execution, the government has formed multimodal committees for port connectivity having representation from major ports (MoPSW), state maritime boards (State & UT Governments) , Ministry of Road Transport and Highways (MoRTH) & Ministry of Railways (MoR) for assessing project wise feasibility, funding arrangement, timelines, implementing agencies etc.

Further under PM Gati Shakti National Master Plan for multimodal connectivity, government has carried out an extensive exercise to find the connectivity gaps between ports and hinterland. Based on the consultation with major ports and state maritime boards, **more than 100 rail and road connectivity projects** have already been identified for implementation in coordination with MoR and MoRTH.

SHIPBUILDING, REPAIR AND RECYCLING

To fulfil the vision of Hon'ble PM towards being Aatmanirbhar Bharat, government has taken multiple steps to promote Shipbuilding, Repair and